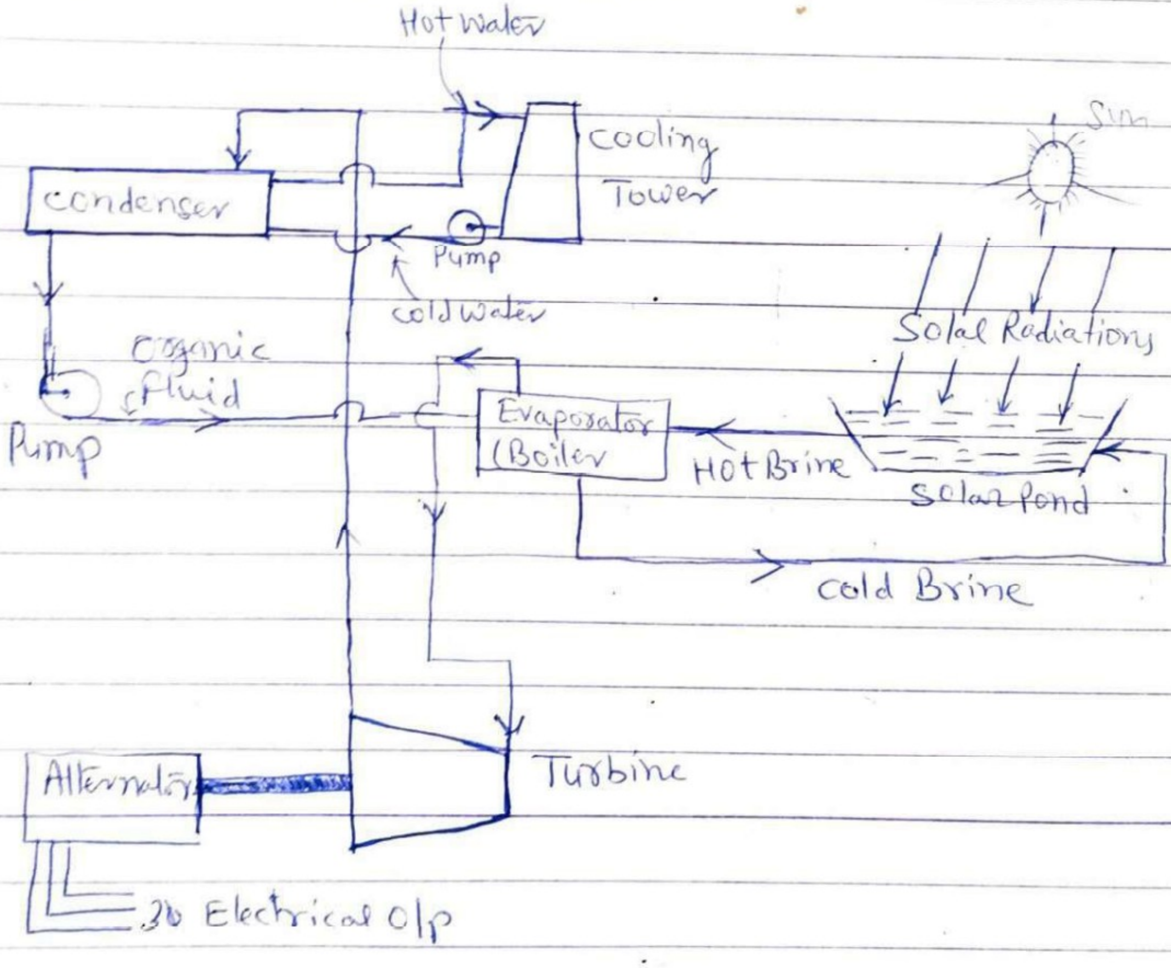
ANS_Q2~~(b)~~

Non conventional sources of energy

(i) Solar energy (ii) Wind energy (iii) Tidal Power
(iv) Geothermal (v) Bio-gas and Bio-Mass (vi) Fuel cells
(vii) Thermo-electric generation (viii) MHD generation
(ix) Thermionic converter

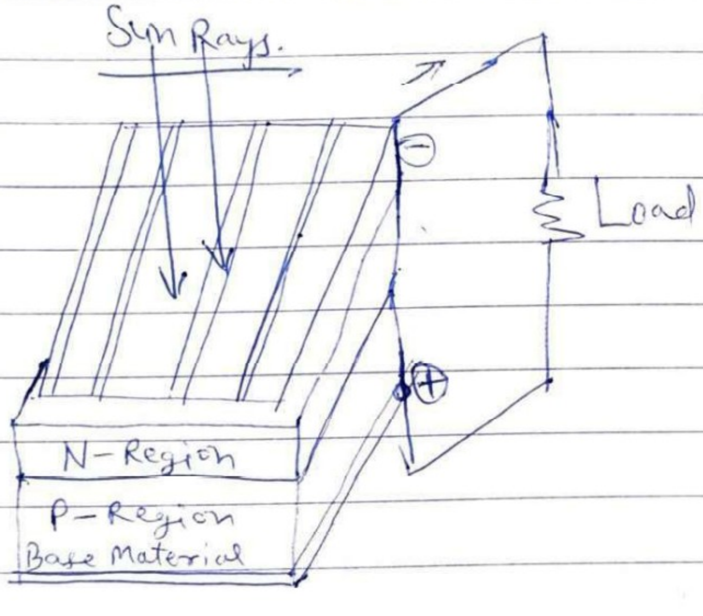
(1) SOLAR Power Plant:



Hot water from Pond is pumped to evaporator, where the organic working fluid to vaporized. The vapour flows under high pressure to the turbine and thereby expanding through Turbine wheel and Electric generator linked to it. The vapour then travels to the condenser where cold water from the cooling tower condense the vapour back it to liquid. The liquid is pumped back to evaporator where the cycle is repeated.

Photo voltaic Power generation →

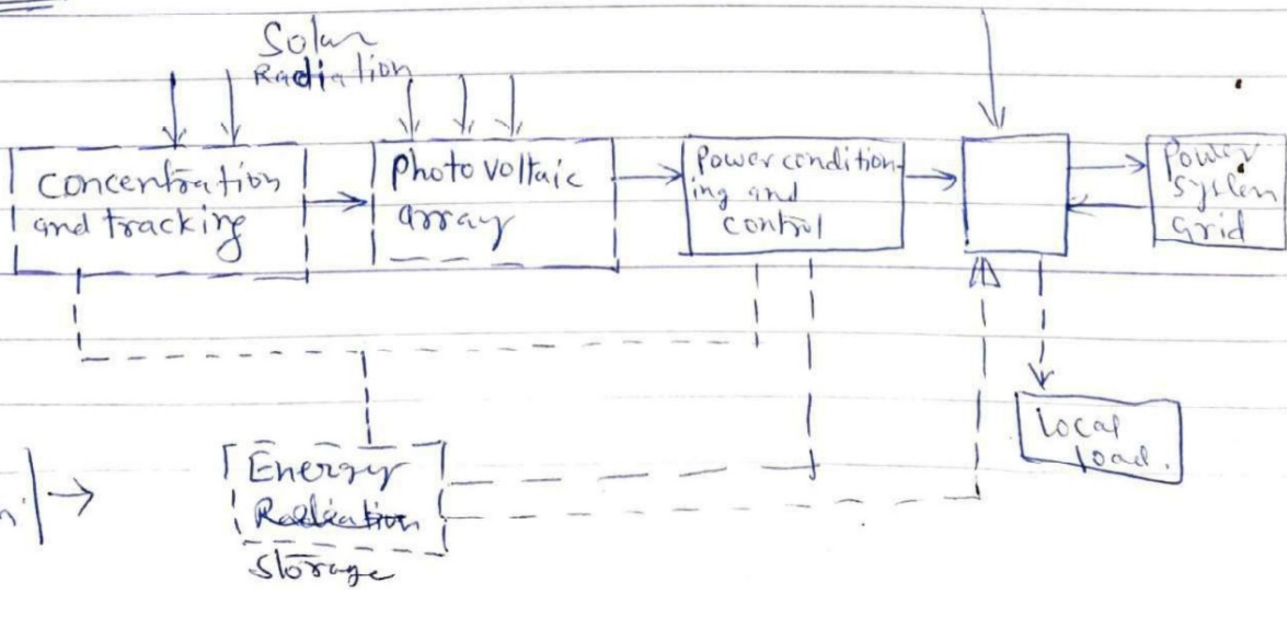
When photons of sun light are absorbed in a semiconductor, they create free electrons (and holes) with higher energy than the electrons which provide the bonding in the base crystals. Once these free electrons-holes pair are created, there must be an electric field to induce these high energy electrons and holes to flow out of the semiconductor to do useful work. It is known that an electric field exist across a p-n junction and this field sweeps the electrons in one direction and holes in the other.



250W/m^2 of array is formed

To ~~dem~~ produce 250MW o/p area required would be 1 Sq. km

Solar cell



Photovoltaic power system →

Advantage of solar cells

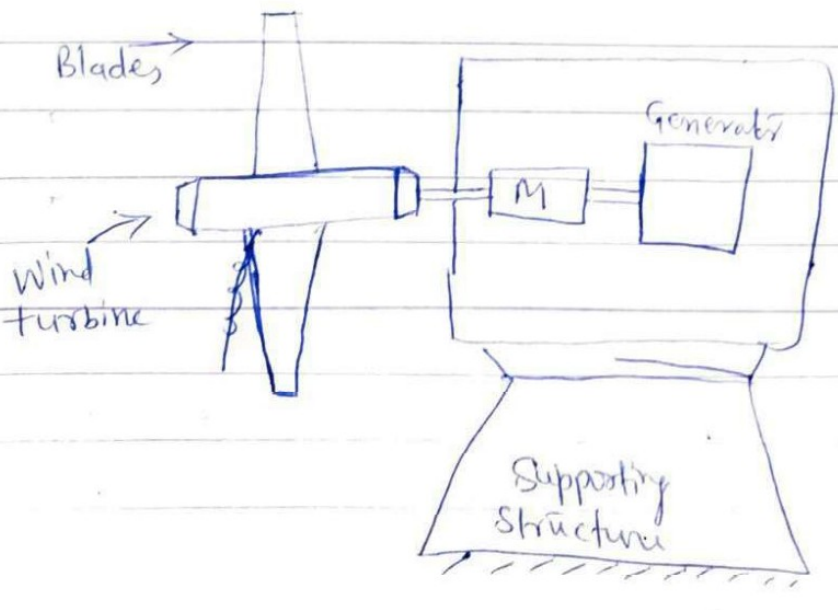
1. need little Maintenance
2. have longer life
3. do not create pollution
4. Energy source is unlimited
5. Easy to fabricate

Disadv.

1. compared to other sources of energy solar cells produce electric power at very high cost.
2. Power o/p is not constant, it Varies with time of day and weather
3. They can be used to generate small amount of Power (Requires large area 250MW - 1 sq km area).

"Wind Mills"

Wind energy is used to run wind mill which in turn drives a generator to produce electricity. Wind mill converts kinetic energy of moving air in to mechanical motion that can be used to run the generator



In India high wind speeds are obtainable in ~~coa~~ coastal areas of Sourashtra. Western Rajasthan, and some parts of central India.

At Present about 43 MW wind power capacity has been established in our country. It is proposed to establish additional 100 MW capacity in the eight five year plan.

Wind Power potential in India is about 20,000 MW.

Advantages: 1) Non Polluting
2) Energy generated is cheaper
3) avoid fuel provision and transport.

Disadv. 1) Wind energy is fluctuating in nature.
2) Because of its irregularity it needs storage devices.
3) There is sufficient noise produced by wind energy Power generating Systems -

Wind Mills in India

1) Madurai Wind Mills at Madurai
2) Sholapur wind Mills at Sholapur
3) Cazri wind mill at Jodhpur (Rajsthan)
4) etc.

Site selection for Wind Mills.

1) Suitable site for wind mill could be found where the annual avergae wind speeds are high about 3.5 - 4.5 m/sec
2) It is desirable to instal Wind Mills at higher altitudes because at higher altitudes wind Velocity is high.

(iii) Ground conditions at the site should be such that the foundations for mills are secured.
The land cost should be low.

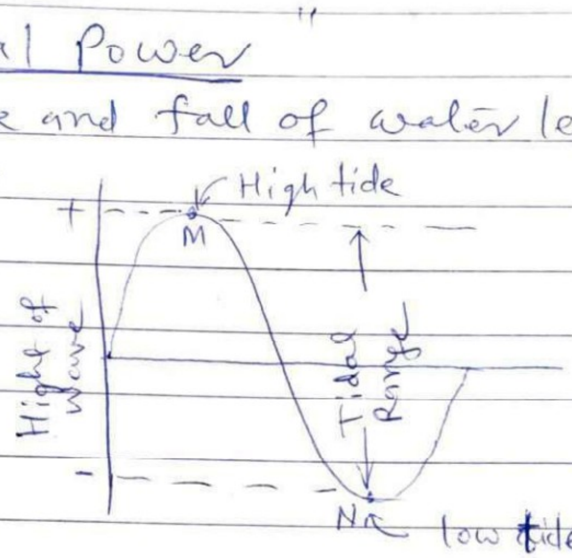
(iv) Icing problems, salt spray or blowing dust should not be present as they affect aeroturbine blades.

(v) Site should be near to users

(vi) Site should be near to road or railway.

"Tidal Power"

Periodic rise and fall of water level in sea is called tide.



Tides occurs due to attraction of sea water by the moon.
These tides can be used to generate electrical power called Tidal power.

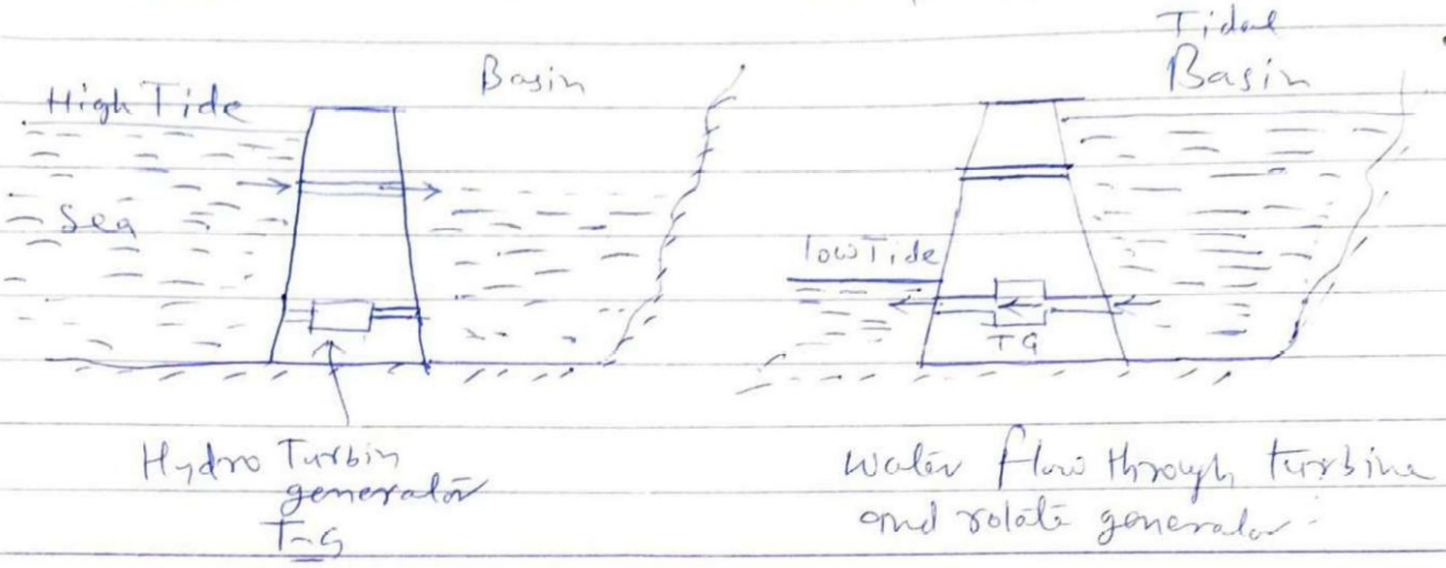
* First Tidal plant was commissioned at 'Rance' in France capacity of 240MW.

Important points for selection of location of Tidal plant :-

1. Tidal range should be adequate throughout the year
2. Site should be free from wave attack of sea.
3. There should be no appreciable change in tidal pattern at the proposed site.

(iv) Site at which tidal power plant is to be located should not have excessive sediment load.

Priciple :- One way system $\rightarrow$ power generated during low Tide only



In two way tidal power plant the power is generated both during flood tide (High tide) as well as during ebb tide (Low tide). The direction of flow through the turbines during ebb and flood tides alternate but m/c acts as a turbine for either direction of flow.

Advantage: (I) free from Pollution
(II) It is inexhaustible and not depends upon rain
(III) Not require large area of Valuable land because they are located on sea shore.
(iv) It has unique capacity to meet Peak power demand when works in combination with thermal station

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Disadvantage → ① o/p varies because of variation in tidal range.
② Transmission cost is high because these are located away from load centres.
③ Sedimentation and silteration are the problems associated with tidal power plants
④ capital cost is high.